Sofia Velasquez Sierra

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Education

Carnegie Mellon University | M.S. Biomedical Engineering (Computational and Machine Learning Track) August 2026

McGill University | B.Eng. Computer Engineering June 2025

Experience

BTG Pactual | Software Engineering Intern | Sao Paulo, Brazil June 2025 - August 2025

- Built 3 internal reporting tools and a data pipeline in React, TypeScript, C#/.NET and PostgreSQL for 15+ credit risk analysts at an investment bank; all 15+ used them daily within 2 weeks and manual reporting time fell 40%
- Agreed requirements with 4 teams before building, and shipped with no scope changes after launch

Human-Computer Interaction Institute, Carnegie Mellon University | Research Assistant May 2026 - Present

- Built a Python data pipeline and automated dashboards that deliver weekly KPI summaries to 2 non-technical stakeholder groups, cutting weekly preparation from about 4 hours to under 20 minutes

Weber Neural Interfaces Lab, Carnegie Mellon University | Research Assistant April 2026 - Present

- Build real-time sEMG decoding pipelines with Meta's engineers and researchers; defined 4 sensor feasibility criteria that Meta adopted to gate hardware decisions

Projects

Glassbox | AI analytics agent that shows its work | glassbox-three-indol.vercel.app 2026

- Built an AI analytics agent where every number is traceable: it states its assumptions as editable values, writes SQL, and a separate Claude call reviews each query (pass, warn or block, with a rewrite) before it runs. Stack: Next.js, TypeScript, Claude API, Supabase Postgres
- Streamed results over typed server-sent events with metrics running in parallel (one question from 87s to 51s); editing an assumption re-runs only the affected queries as signed SQL with no model call
- Locked down SQL execution in layers: a Postgres parser enforcing a single SELECT with allowlists, an EXPLAIN cost ceiling, a read-only transaction with a timeout, and a least-privilege role checked by 24 permission tests
- Designed a token-based light and dark design system with a colour-blind-validated chart palette and a documented /system page; 94 unit tests and zero axe violations; reported a 10-question golden-set eval honestly at 7/10

Rollout | Counterfactual debugger for AI agents | Next.js, TypeScript, three.js, Claude API October 2026

- Built a tool to scrub, fork and compare AI agent runs in a simulated 3D world; Claude Haiku drives the agent and Claude Sonnet scores each branch against a fixed rubric
- Deterministic event-log simulator where a fork is a pointer, not a copy, with 68 tests; four full design iterations, keyboard control and reduced-motion support

AgentTrace | AI agent evaluation tool | Python, Claude API May 2026

- Built a streaming chat tool in which one Claude call plays the agent and a second scores the conversation, returning 3 concrete fixes per run

ROF | Web platform | TypeScript, React, Next.js, tRPC, Prisma, Tailwind December 2024

- Designed and shipped a 9-page platform with onboarding, ID verification and an analytics dashboard for real users, owning the front end, API layer and database schema

Skills

Front end: TypeScript, React, Next.js, Tailwind, CSS, SVG data visualization, design tokens and component systems, accessibility (axe), animation
AI and data: Claude API, streaming (SSE), agent design and evaluation, PostgreSQL, SQL safety, Supabase, Python, Claude Code, Cursor
Also: Node.js, tRPC, Prisma, Git, Vercel, Figma; English, French, Spanish, Portuguese